🎓 Student Dropout & Academic Success Analysis

End-to-End Analytics & Dashboard Portfolio Project Blueprint

Dataset: Kaggle | **Rows: 4,424** | **Columns: 35** | **Tools: Python, Pandas, Power BI / Excel**

📌 1. Project Overview & Motivation

This project aims to analyze the key factors influencing higher education student outcomes—specifically identifying drivers behind student dropouts versus academic completion. The goal is to produce actionable data-driven insights and an interactive dashboard for academic decision-makers.

4,424	32.1%	49.9%
TOTAL STUDENTS	OVERALL DROPOUT RATE	GRADUATE RATE

📊 2. Baseline Findings (Exploratory Data Analysis)

Preliminary exploratory analysis using Python/Pandas revealed critical risk factors strongly correlated with student attrition:

Risk Factor Category	Segment Group	Dropout Rate	Key Takeaway
Scholarship Status	No Scholarship Holder	**38.7%**	Scholarship recipients are >3x less likely to drop out.
	Scholarship Holder	**12.2%**	
Tuition Fees Status	Not Up to Date (Defaulters)	**86.6%**	Fee status is the strongest early warning indicator of dropout.
	Up to Date	**24.7%**	

🖥️ 3. Interactive Dashboard Requirements (Power BI / Excel)

Page 1: Executive Overview

- ☐ **KPI Cards:** Total Student Count (4,424) | Overall Dropout Rate (32.1%)
- ☐ **Donut/Pie Chart:** Target Distribution Breakdown (Graduate: 49.9%, Dropout: 32.1%, Enrolled: 18.0%)

Page 2: Risk Factor Analysis

- ☐ **Bar Chart:** Dropout Rate % by Scholarship Holder (Yes vs No)
- ☐ **Bar Chart:** Dropout Rate % by Tuition Fee Status (Up to Date vs Pending)
- ☐ **Bar Chart:** Dropout Rate % by Gender
- ☐ **Bar Chart:** Dropout Rate % by Debtor Status (Financial liability flag)

Page 3: Academic Performance Trends

- ☐ **Scatter / Bar Chart:** Average *Curricular units 2nd sem (approved)* grouped by Target status.
- ☐ **Comparison Chart:** 1st Semester Average Grade comparison across Graduates vs Dropouts.

🔍 4. Exploratory Tasks (Python Inquiries to Conduct)

Perform initial grouping and crosstabs in Python to discover 1-2 additional high-impact insights for the dashboard and resume:

- **Age at Enrollment:** Bin into age brackets (`<20`, `20-25`, `25+`) to evaluate mature student risk.
- **Displaced Status:** Test if relocated students exhibit significantly different dropout rates.
- **Parental Education Level:** Measure correlation between Mother's/Father's qualifications and student success.
- **Application Mode / Order:** Evaluate if application stream correlates with persistence.

📋 5. Final Deliverables Checklist

- ☐ Power BI Report File (`student_analysis.pbix`) containing 3 configured visual pages.
- ☐ 1-2 High-resolution exported screenshots (PNG) of the final dashboard pages.
- ☐ Cleaned Jupyter Notebook (`.ipynb`) containing Pandas data cleaning and analysis pipeline.
- ☐ Updated Resume Bullet Points summarizing 2 key business insights (Scholarship & Fee Defaulter findings).

> ### ⚠️ Execution Rules & Constraints
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> - **Attendance Column Usage:** Do NOT label or interpret the `Daytime/evening attendance` column as attendance percentage. It refers strictly to class time shift.
> - **Data Integrity:** Do NOT round or artificially alter percentage metrics. Preserve exact figures.